

Focus on Research Methods

Real Qualitative Researchers Do Not Count: The Use of Numbers in Qualitative Research

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Received 24 August 2000; accepted 12 January 2001

Abstract: Two myths about qualitative research are that real qualitative researchers do not count and cannot count. These antinumber myths have led to the underutilization of numbers in qualitative research and to the simplistic view of qualitative research as non- or antinumber. Yet numbers are integral to qualitative research, as meaning depends, in part, on number. As in quantitative research, numbers are used in qualitative research to establish the significance of a research project, to document what is known about a problem, and to describe a sample. But they are also useful for showcasing the labor and complexity of qualitative work and to generate meaning from qualitative data; to document, verify, and test researcher interpretations or conclusions; and to re-present target events and experiences. Although numbers are important in the treatment of qualitative data, qualitative researchers should avoid the counting pitfalls of verbal counting, over-counting, misleading counting, and acontextual counting. © 2001 John Wiley & Sons, Inc. Res Nurs Health 24:230–240, 2001

Keywords: qualitative research; numbers; counting; analysis; re-presentation

One myth about qualitative research is that real qualitative researchers do not count. Contained in this myth are the depiction of words and numbers as two different and opposing entities that cannot easily or “shamelessly” coexist without considerable effort (Ford-Gilboe, Campbell, & Berman, 1995; Rossman & Wilson, 1994) and the equating of qualitative research with words and qualities and of quantitative research with numbers and quantities (Bauer, Gaskell, & Allum, 2000). Indeed, qualitative research is often defined by the absence of numbers, in addition to a critique of numbers, in which they are associated with the putatively superficial preoccupations of post- or neopositivist researchers with *how much* and *how many*. At the very least, numbers constitute something of a “litmus test” (Linnekin, 1987,

p. 920), differentiating scientifically oriented (or quantitatively informed) from humanistically oriented qualitative research (Chibnik, 1999).

A second and related myth is that qualitative researchers cannot count. Contained in this myth is the depiction of less competent researchers (that is, not competent in numbers) as turning to qualitative research to avoid the rigors of quantitative research, as well as a dichotomous view of verbal and mathematical skill as being either/or. I have been amused at the responses from colleagues who seem amazed at my ability to add, subtract, multiply, and divide. The very label *qualitative researcher* seems to imply a deficit in mathematical abilities; in contrast, the label *quantitative researcher* seems to imply no comparable deficit in verbal abilities.

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These antinumber myths have led to the under-recognition, underutilization, and even avoidance of numbers in qualitative research. Although they have a less prominent place in qualitative research, numbers are integral to qualitative data, and skill with numbers is essential to good qualitative research. The meaning qualitative researchers seek depends, in part, on number, just as number depends on meaning (Dey, 1993). As in quantitative research, numbers (e.g., in the form of demographic, epidemiologic, and survey data) are used in qualitative research to establish the significance of a research problem, to document what is known about a problem, and to describe a sample. Numbers (e.g., in the form of statistical findings) are also used in combination with qualitative data in mixed method studies. In addition, numbers are powerful rhetorical devices (John, 1992) that can be used to showcase the labor and complexity of qualitative work. For example, instead of apologizing for the so-called limitations of their "small" sample sizes, researchers might show the "large" numbers of which such ostensibly small samples are often actually comprised. A former doctoral student of mine (Bailey, 2000) recently completed a 17-month ethnographic study of the implementation of one computerized patient record system on one hospital unit that included interviews with 20 informants totaling 325 pages of text, 124 observations of events captured in 1162 pages of field notes, and the review of documents totaling 820 pages of text. A study of 10 participants, interviewed only once, can yield 250 pages of raw data alone. Qualitative researchers can take advantage of the rhetorical power of numbers and thereby show the true sample size and amount of data produced from even $N = 1$ studies.

In this article I focus on the uses of numbers in analyzing, interpreting, and re-presenting qualitative data in primarily qualitative studies. Specifically, I consider the use of numbers to generate meaning from qualitative data; to document, verify, and test researcher interpretations or conclusions; and to re-present target events and experiences. These uses of numbers are not mutually exclusive. That is, although numbers may be used primarily to generate meaning, they will inevitably also serve to document the steps taken to arrive at meanings and then to verify them. Similarly, numbers intended primarily to document and verify will also inevitably enable researchers to extract more meaning from qualitative data.

GENERATING MEANING

Counting is integral to the analysis process, especially to the recognition of patterns in data and deviations from those patterns, and to making analytic or idiographic generalizations from data. Pattern recognition implies seeing something over and over again in one case or across a selection of cases. Finding that a *few*, *some*, or *many* participants showed a certain pattern, or that a pattern was *common*, *thematic*, or *unusual* in a group of participants, implies something about the frequency, typicality, or even intensity of an event. Such counting is often unconscious, as qualitative researchers strive to emphasize something more than the numbered nature and meaning of events and experiences. Yet anytime qualitative researchers take raw data and place it into categories or discern regularities in it to which they attach codes or designate as themes, they are drawing from the numbered nature of phenomena for their analysis. At the very least these data may be subject to scoring as nominal data, as researchers name and consider the presence or absence of events, experiences, or themes.

Accordingly, depending on sample size and composition, descriptive but also inferential statistical measures can be used to extract more meaning from a qualitative data set, as certain of these data sets will lend themselves to transformation or translation into quantitative data sets. Tashakkori and Teddlie (1998, p. 126) referred to this process as "quantitizing," while Boyatzis (1998, p. 129) referred to it as "quantitative translation." Here qualitative "themes" are numerically represented in scores, scales, or clusters in order to more fully describe and/or interpret a target phenomenon. For example, Blaxter (1983) listed the number of instances in order of prevalence that 46 women gave to 11 categories of causes of conditions or diseases. Most prevalent was the category of infection, and least prevalent was the category of aging. In each of the 11 categories Blaxter listed the most frequently mentioned conditions and the numbers of women naming them.

Displaying information numerically can make patterns "emerge with greater clarity" (Dey, 1993, p. 198). In Blaxter's study the reader learns from her discussion of the findings and sees in her tabular displays that women attributed more than one cause to the same condition. For example, tuberculosis appears as a condition attributed to infection, family tendencies, environmental poisons, trauma consequence, and a neglect and poverty consequence. Tuberculosis was the most

frequently mentioned disease ($n = 36$), but it also ranked second among the diseases most frequently mentioned without discussion of any cause ($n = 16$). As a result of having counted the women, diseases, and causes in her study, Blaxter discerned a finding worthy of interpretive effort. Tuberculosis seemed to be such an important disease, and yet it was perceived as so common, even so trivial, that it warranted both discussion and no discussion of its cause.

In a study a colleague and I conducted with women and couples obtaining positive diagnoses of fetal impairments during pregnancy (Sandelowski & Jones, 1996a), a numerical display suggested that such patients might be as or even more usefully differentiated theoretically by the stories they told about deciding whether to continue or terminate their pregnancies as they were differentiated empirically by the choices they actually made and the “factual” circumstances surrounding those choices. As shown in Table 1, women and couples in the same circumstances (having either a viable or lethally impaired fetus with and without the option to terminate their pregnancies) and deciding on the same course of action (if they had the option to continue or to terminate their pregnancies) also told the same stories of choosing. Yet there were exceptions in one empirical category and one theoretical category. That is, the “lost choice” story (a theoretical category) appeared in two groups of participants (empirical categories), while both “lost choice” and “found choice” stories appeared in one group of participants. The sample size and composition for the whole study were inadequate either qualitatively or quantitatively to verify any narrative pattern but large enough to generate questions for future inquiry

concerning the range of story types available to people experiencing common adverse life events and their impact on well-being. Indeed, the very numerical display of this result is what led us to see that narrative groupings did not always coincide with empirical ones and to further explore the functions these stories served.

Accordingly, number play can be useful in searching for patterns and for generating working hypotheses. Tables 2, 3, and 4 show some number play to illuminate the extent to which knowing during pregnancy that a fetus is impaired adds to the burden of having an impaired child. Five couples had suggested to us that having foreknowledge interfered with the pleasure of pregnancy (Sandelowski & Jones, 1996b). Consequently, I decided to calculate the length of time couples had available to them to assume everything was all right and the time they spent in pregnancy knowing something was wrong in order to ascertain whether the durations of the pregnancy spent with and without this knowledge differentiated couples in ways that might ultimately prove to be analytically or clinically relevant. That is, was there any suggestive pattern that might emerge by creating a visual display containing information for each couple on when they learned they were expecting a baby, when they learned that baby was impaired, and when their pregnancy ended? Could time spent with and without foreknowledge be used as a primary organizing principle to place couples in groups that might later be linked to outcomes, such as narrative strategies or parenting styles?

As shown in Table 2, I created a data display from those interview transcripts that contained all information required to compute how much of the pregnancy couples had with and without knowl-

Table 1. Circumstances and Stories of Choosing in Couples With Positive Prenatal Diagnoses

Circumstances	Stories				
	Nature	Disowned	Lost	Close	Found
Viable fetus + option ($n = 5$)					
Continued		3			
Terminated				2	
Nonviable fetus + option ($n = 5$)					
Continued					
Terminated	5				
Viable fetus – option ($n = 3$)			3		
Nonviable fetus – option ($n = 2$)			1		1

Note. From “‘Healing fictions’: Stories of Choosing in the Aftermath of the Detection of Fetal Anomalies,” by M. Sandelowski and L. C. Jones 1996, *Social Science & Medicine*, 42, p. 357. Copyright 1996 by Elsevier Science. Adapted with permission.

Table 2. Display of Information Required to Discern Duration of Knowledge Burden in Pregnancy

Cases	GA ^a at Pregnancy dx	GA at Impairment dx	GA at Birth/Death/ Termination	Without Knowledge	With Knowledge
1	10	30	38/birth	20	8
2	11	30	37/birth	19	7
3	4	21	22/termination ^b	17	1
4	4	16	20/termination	12	4
5	7	26	37/birth	19	11
6	6	18	36/birth	12	18
7	4	18	19/termination	14	1
8	4	35	40/birth	31	5
9	1	16	16/termination	15	<1
10	4	17	22/termination	13	5
11	4	15	19/termination	11	4
12	4	18	19/termination	14	1

^aAll gestational ages (GAs) and time periods in weeks.

^bTermination after fetal demise.

edge of fetal impairments. Playing this number game not only enabled me to see my data in a new way (and therefore potentially to see new things in my data), but also to see more about the characteristics of my sample. As shown in Table 2, the 7 couples who ultimately terminated their pregnancies had received positive fetal diagnoses earlier in pregnancy (as early as 15 weeks and no later than 21 weeks) than the 5 couples who ultimately continued their pregnancies (as late as 35 weeks and no earlier than 18 weeks). As shown in Tables 2 and 3, among the 7 couples terminating, no couple spent more than 5 weeks with the knowledge that something was wrong with their baby; 4 of these couples spent only 1–2 weeks with foreknowledge as they terminated their pregnancies within 1 week of obtaining a positive fetal diagnosis. Among the 5 couples continuing pregnancies, 3 spent at least 7 weeks with foreknowledge, with 1 couple in this state for 18 weeks. As shown in Tables 2 and 4, of the 7 couples terminating their pregnancies, 6 couples had no more than 15 weeks of pregnancy without foreknowledge. In contrast, of the 5

couples continuing their pregnancies, 4 had at least 17 weeks without foreknowledge. Among the questions raised by this number play was whether a greater time in pregnancy not knowing about a fetal impairment generates a more favorable response to a child, or whether a greater time knowing helps parents adjust more quickly to such a child after birth. What is more important to the ultimate well-being of parents: when they find out or how long they know about a fetal impairment?

Not only can displaying information numerically make patterns emerge more clearly or, at the very least, generate new questions or new lines of analysis, it can also clarify meaning. Sometimes data displays are so overloaded with words or verbal explanations are so complex that the reader will have difficulty absorbing all the information conveyed in them. Reducing qualitative data to numbers can sharpen the focus on a key finding. Cohen, Tripp-Reimer, Smith, Sorofman, and Lively (1994) analyzed professionals' and patients' explanations of diabetes by comparing them on five parameters: etiology,

Table 3. Period of Foreknowledge in Couples Continuing and Terminating Pregnancies With Fetal Impairments (*N* = 12)

Weeks Knowing	Continuing (<i>n</i> = 5)	Terminating (<i>n</i> = 7)
1–2	0	4
4–5	1	3
7–8	2	0
11	1	0
18	1	0

Table 4. Period of No Knowledge in Couples Continuing and Terminating Pregnancies With Fetal Impairments (*N* = 12)

Weeks Not Knowing	Continuing (<i>n</i> = 5)	Terminating (<i>n</i> = 7)
11–13	1	3
14–15	0	3
17–20	3	1
31	1	0

time and mode of symptom onset, pathophysiology, severity, and treatment. In the body of their report they verbally detailed participants' responses and indicated important areas of disagreement existed. They then referred the reader to a table (Table 5) showing the number of professional/patient partners agreeing, slightly disagreeing, and largely disagreeing about the etiology, pathophysiology, severity, and treatment of diabetes. By restricting this table to a numerical representation of findings, instead of loading it with qualitative detail, these researchers enabled readers to see at a glance where the areas of agreement and disagreement clustered. Moreover, they counted only the "countable" (Seale, 1999, p. 121). Not all qualitative data (and some would argue, very little) are amenable to counting. Although numerical comparisons of time and mode of symptom onset were not made because professionals' responses were too general to count, these responses were not ignored; they were verbally detailed.

Cohen et al. (1994) also computed total congruence scores for each professional/patient-partnership and then used the Spearman correlation coefficient to ascertain whether any significant relationships existed between these congruence scores and glycosylated hemoglobin levels. They found a positive ($r=0.267$) but nonsignificant ($p=0.355$) relationship between explanatory model congruence and normal hemoglobin A1c levels. They also discerned a "noteworthy . . . trend" (Cohen et al., p. 64), in that of those patients who were in agreement with their professional partners, 68% had normal hemoglobin A1c levels, while 52% of patients with a minor disagreement and 50% of patients with a major disagreement had normal levels. This ethnographic study illustrates the varied and insightful use of quantitative analysis techniques

to extract meaning from qualitative data. The "composite model of diabetes" (p. 64) they created was based on fine ethnographic detail derived in part from the various calculations they made (only several of which I feature here). Indeed, such use of numbers has long been an integral part of the ethnographer's toolkit (LeCompte & Schensul, 1999; Ryan & Bernard, 2000).

DOCUMENTING, VERIFYING, AND TESTING CONCLUSIONS

Counting is essential to ensuring descriptive (getting the "facts" of the case right), interpretive (getting participants' interpretations of events right), and/or theoretical (developing an interpretation that fits these facts) validity (Maxwell, 1992). Counting assures the reader that researchers have accounted for all and have not "discount[ed]" (Dey, 1993, p. 220) any of their data, thereby avoiding the major pitfalls in qualitative analysis. These pitfalls include: (a) overweighting dramatic or vivid accounts of events; (b) underweighting data that do not conform to the pattern the researcher wants to find (also called the "holistic fallacy" [Miles & Huberman, 1994, p. 263]); and, (c) regressing to the mean—smoothing out or averaging out—or otherwise cleaning up the contradictions and messiness of human accounts and lives. "Playing the numbers game" (Dey, 1993, p. 223) can be a useful corrective and is an element of the audit trail by which researchers can document their procedural and analytic moves (Rodgers & Cowles, 1993).

As shown in Figure 1, Koenig (1988) enumerated the case reports and articles featuring therapeutic plasma exchange over time to justify

Table 5. Use of Numbers to Reduce Key Finding

Parameter	Agreement Between Professionals' and Patients' Explanations (N=14)		
	Agree	Minor Discrepancy	Major Discrepancy
Etiology	2	5	7
Pathophysiology	3	1	10
Severity	6	6	2
Treatment	8	5	1

Note. From "Explanatory Models of Diabetes: Patient Practitioner Variation," by M. Z. Cohen, T. Tripp-Reimer, C. Smith, B. Sorofman, and S. Lively, 1993, *Social Science & Medicine*, 38, p. 61. Copyright 1993 by Elsevier Science. Adapted with permission.

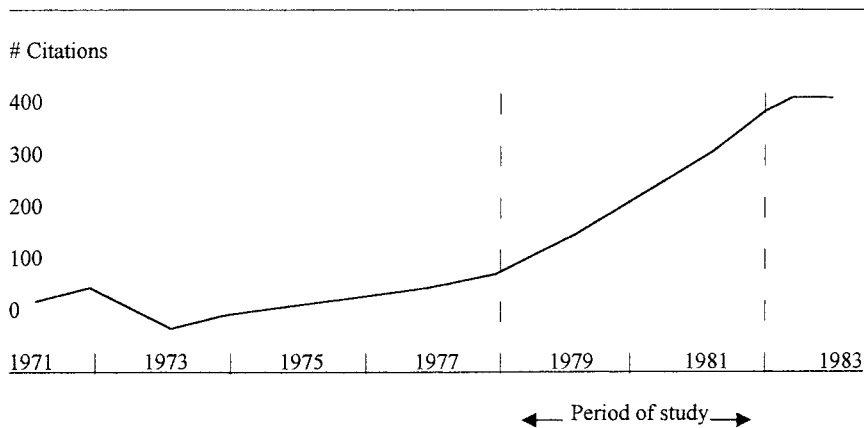


FIGURE 1. Number display to make the case for a case. *Note.* From "The technological imperative in medical practice: The social creation of a 'routine' treatment," by B.A. Koenig, in *Biomedicine Examined* (p. 472), edited by M. Lock and D. Gordon, 1988, Dordrecht, The Netherlands: Kluwer Academic. Copyright 1988 by Kluwer Academic. Adapted with kind permission of the publisher.

choosing this technology to exemplify the rapid move from experimental to routine treatment. In order to study the process of routinization, Koenig had to make the case that she had in fact chosen a case (that is, a therapeutic innovation) that would permit her to study this process ethnographically. As shown in Figures 2 and 3, Borkan, Quirk, and Sullivan (1991) used numerical displays to document and verify both the procedural and analytic moves in the qualitative portion of their mixed method study with elders with hip fractures. Included in Figure 2 are accounts of "no data," a common and expected occurrence in qualitative research, as the objective in data collection is not primarily to collect comparable data sets, but rather data sets that are complete for each case. The numerical displays in Figure 3 were important to establish the process by which these researchers ensured an accepted level of interrater reliability, as their ultimate goal was to link narratives quantitatively to functional outcomes.

An example of the use of numbers primarily to confirm an impression and only secondarily to more fully to describe a sample is illustrated in Table 6. I had the impression that the infertile couples I observed had more experience with adversity, both because of and beyond their infertility, than the fertile couples observed. I listed and enumerated all the events couples mentioned having experienced and which are commonly viewed in North American culture as adverse. After organizing this information by

fertility group, I was able to confirm my impression that as a group the infertile couples had suffered not only more reproductively related adversity, but also more divorces and serious illnesses, than the fertile couples.

An example of a schematic rendering of numbered data to support an interpretation is the display by Simon (1999, p. 148) of the spatial dynamics during a typical diagnostic imaging conference. As shown in Figure 4, Simon enumerated the women and men by shading the male participants and onlookers in order to lend more credence to his conclusion that the spatial dynamics in these conferences reflected prevailing gender dynamics.

RE-PRESENTING DATA AND LIVES

Numbers also are used for re-presenting events in qualitative research, as when means and frequencies are used to describe a sample in the body of the text itself or in a visual display such as a table, graph, or schematic drawing. Qualitative researchers in the health sciences often want to satisfy both scientific and humanistic/artistic criteria in re-presenting the lives they have studied. They want to convey procedural and analytic rigor, but they want to do this in ways that will not undermine the aesthetic qualities of the report itself. Qualitative researchers tend to prefer figures of speech over figures, and tableaux of experience over tables of numbers. Numbers

Poles: Organic	1-3	4	5-7	Mechanistic	No data
Name					
	Disease			Fracture	
	1%	15%	49%		35%
Illustrative quotes:					
	"sickness"			"broken hip"	
	"osteoporosis"			"shattered bone"	
Pathophysiology					
	Broke and fell			Fell and broke	
	10%	14%	64%		12%
Illustrative quotes:					
	"I think you break your hip before you go down."			"I landed on my hip and knew it was broke."	
Prognosis					
	Total impairment			Complete recovery	
	14%	10%	70%		6%
Illustrative quotes:					
	"I might have to go to a nursing home."			"I'm going to recover fully."	

FIGURE 2. Numerical display to document analytic moves. *Note.* From "Finding Meaning After the Fall: Injury Narratives from Elderly Hip Fracture Patients," by J. M. Borkan, M. Quirk, and M. Sullivan, 1991, *Social Science & Medicine*, 33, p. 951. Copyright 1991 by Elsevier Science. Excerpted and adapted with permission.

illuminate but also obscure and thereby complicate the necessary tension between science and art in qualitative research. The four complications comprising the “dubious use” of numbers (Stern, 1989, p. 139) are verbal counting, overcounting, misleading counting, and acontextual counting.

Verbal Counting

Verbal counting occurs when researchers imply numbers without actually giving any. For example, researchers will use such words as *a few*,

some, *many*, and *most* to refer to participants, *common*, *prevalent*, *sometimes*, and *rare* to refer to themes, *thematic* to refer to experiences, and other such quantitative designations to qualify findings without telling the reader what they mean in the context of a specific research report. Here, word masks number. Writers have several options. They can use actual numbers in place of pronouns connoting indeterminate quantity. For example, they can state that *3 women believed that tuberculosis was caused by an infection* or *all but 3 of the women believed tuberculosis was caused by poor diet*. Writers can also choose to

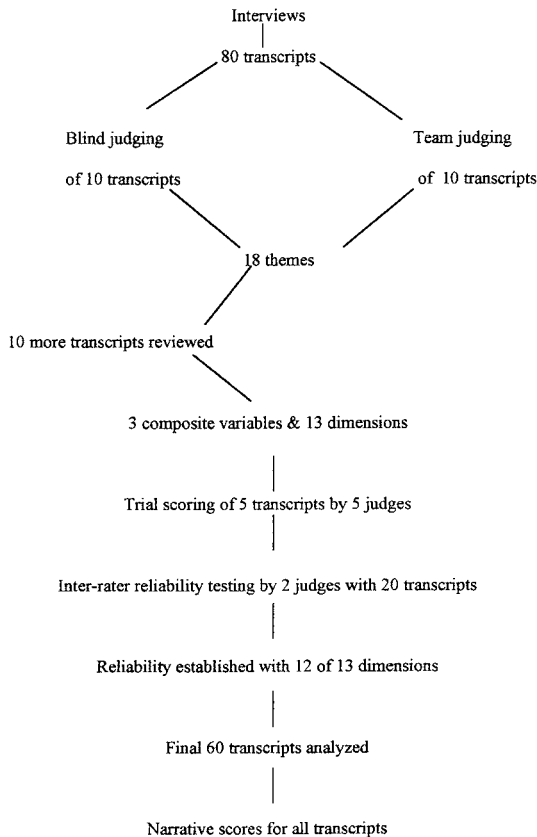


FIGURE 3. Numerical display to document procedural moves. *Note.* From "Finding Meaning After the Fall: Injury Narratives from Elderly Hip Fracture Patients," by J. M. Borkan, M. Quirk, and M. Sullivan, 1991, *Social Science & Medicine*, 33, p. 950. Copyright 1991 by Elsevier Science. Adapted with permission.

Table 6. Number Display to Confirm Impression

Adverse Life Events	Fertile (N = 19)	Infertile (N = 75)
Divorce	4	24
Widowhood	0	1
Pregnancy/infant loss	2	17
Elective termination	3	3
Adoption failure	0	8
Serious illness	1	11

Note. From *With Child in Mind: Studies of the Personal Encounter with Infertility* (p. 262), by M. Sandelowski, 1993, Philadelphia: University of Pennsylvania Press. Copyright 1993 by University of Pennsylvania Press. Adapted with permission.

use pronouns connoting indeterminate quantity but first advise readers (in the body of the report or in a legend or footnote) what their pronouns connoting indeterminate quantity will mean. For example, the writer can operationally define *common* as something occurring in more than 50% of participants and *few* or *rarely* as occurring in less than 20% of participants.

Overcounting

The danger here is that writers, eager to offset the vagueness of verbal counting, will become so preoccupied with providing exact numbers that they will end up overcounting. There are two kinds of overcounting: representational and analytic.

Representational overcounting. In this first and less serious type of overcounting, researchers count what should be counted, but overuse numbers in ways that detract from the aesthetic presentation of findings. Accordingly, writers will want to avoid the *3 women said this . . . 6 women said that . . . 10 women said something else* motif. Ways to avoid this kind of overcounting include the type of operationally defined verbal counting mentioned previously or the use of data displays showing frequencies. Seals et al. (1995) used three tables to indicate the number of responses each of their focus-group questions elicited from women and the number of responses in each category of social service concerns or issues they discerned. This allowed them to focus on the qualitative nature of these responses in the main body of the text. When they indicated that "most commonly, women described the inconveniences of using social services" (p. 505), the reader could look at their tables to see what "most commonly" means.

Writers can also avoid representational overcounting by adding relevant descriptive material concerning participants and by varying the style of presentation. For example, a writer might report that *the 3 women who believed the disorder was caused by an infection all had relatives who first had an infection and then developed the disorder*. The idea here is to integrate the use of numbers into the presentation of findings and thereby avoid having numbers take over the presentation.

Analytic overcounting. More serious forms of overcounting include counting just for the sake of counting and counting the uncountable. Here, qualitative researchers use numbers in ways that divert them from their primary mandate of

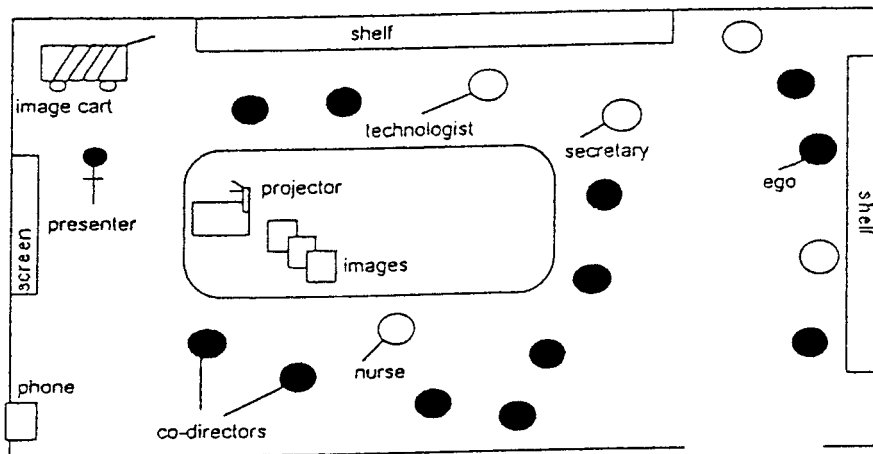


FIGURE 4. Numbered schematic display of spatial dynamics. *Note.* From "Images and Image: Technology and the Social Politics of Revealing Disorder in a North American Hospital," by C. M. Simon, 1999, *Medical Anthropology Quarterly*, 13, p. 148. Copyright 1999 by American Anthropological Association. Reprinted with permission.

developing and presenting a fully rounded interpretation. Analysts may be so eager to be numerically precise they end up counting everything that can conceivably be counted, even though this will add nothing to the understanding of a target phenomenon. For example, some, but not all, types of qualitative studies require that analysts count how many themes they discerned in their data. In the Borkan et al. study (1991) mentioned previously, enumerating the themes they discerned was critical to documenting how they transformed a large volume of verbal data into two variables representing two narrative explanations of hip fracture. The treatment of the narrative data in this study was quantitatively informed; counting themes here was not overcounting.

In contrast, an example of analytic overcounting is emphasizing the numbers of respondents falling into a category instead of the qualitative nature of the categories themselves in reports by researchers conducting grounded-theory work. If an analyst finds, for example, five orientations to the future exhibited in a sample of 25 women with HIV, the main thrust of the research report is to persuade the reader that there were different and mutually exclusive patterns of response, which the analyst has labeled future orientations. The analyst will compare each of these five orientations on such parameters as the defining attributes of the orientation, the women most likely to show it, and the consequences of having each orientation. Documenting that of the 25 women, 5 showed Pattern (or orientation) A, 10

showed Pattern B, 7 showed Pattern C, 1 showed Pattern D, and 2 showed Pattern E is simply to document that analysts have accounted for all the participants. Analysts might refer to Pattern B as the prevalent one but must add that it is such in this group of women only. Moreover, the prevalence of any one pattern in a purposefully selected sample is less relevant than the discovery of five mutually exclusive orientations to the future, as such samples do not allow analysts to make inferences about prevalence beyond their sample. In addition, although the analyst will note that there were five patterns discerned, that there were five, as opposed to four or six, patterns has less importance than detailing the patterns themselves. Accordingly, the key findings in the study by Timmermans (1997) of resuscitation in hospital emergency rooms are the "survival," "bifurcated," and "holistic" (p. 157) resuscitation perspectives he discerned from interviews and observations. How many resuscitation events or health care providers fell into each of these theoretical groupings is less relevant to the purpose of his report (and therefore may or may not be reported) than his ability to persuade the reader that he has in fact identified different resuscitation perspectives.

Misleading Counting

In providing actual numbers writers will want to avoid counting that misleads readers. For example, writers should avoid using percentages to

describe small samples (but some journals may require this mode of numerical re-presentation). In a sample of two, it is highly misleading to talk about 50% of a sample doing this and 100% of a sample showing that. A good rule of thumb is that if a total sample involved less than 25 cases (e.g., persons, families, organizations), it is more accurate to give the actual number showing one thing or another.

Acontextual Counting

Finally, writers will want to avoid acontextual counting whereby they offer no other information about a participant or event except numbers or in which they draw unsubstantiated inferences from numbers. One example of acontextual counting is when researchers say that 40% of the children they observed were angry but offer no other information about these children. Another example is when researchers count up the number of times persons referred to a certain event and then conclude that this event was the most important event in the lives of those persons who referred to it the most. Or they count up the number of times participants used the word *angry* and conclude that people who used it more frequently were angrier than those who used it less frequently. Without some narrative or other theoretical formulation concerning talk or the salience of life events that would allow researchers to draw such conclusions, the conclusions remain theoretically invalid.

CONCLUSION

The desire to move “beyond numbers” (Greenhalgh & Taylor, 1997) should not prevent qualitative researchers from using numbers to get there. Numbers can complement and enhance narratives (Olson, 2000). But qualitative researchers are challenged to use numbers in ways that produce both trustworthy findings and evocative reports of those findings. They must decide when to count and what is countable in order to make their use of numbers count.

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